1 Rearranges to collect x terms on one side and constants on the other, e.g., $-8 < 4x$ or $-4x < 8$ **[M1]**

$x > -2$ or $-2 < x$ **[A1]**

[Total Marks: 2]

2 **(a)** $3x - 6 > 5x + 4$ **[M1]**

$x < -5$ **[A1]**

(b) -6 **[B1]**

[Total Marks: 3]

3 Expands brackets correctly: $12x - 4 - 5x \geq 2x + 16$ **[M1]**

Rearranges to collect x terms: $7x - 2x \geq 16 + 4$ or $5x \geq 20$ **[M1]**

$x \geq 4$ **[A1]**

[Total Marks: 3]

4 **(a)** $3x - 12 < 5x - 2$

[M1]

$-10 < 2x \implies x > -5$ (or $-5 < x$)

[A1]

(b) -4

[B1]

[Total Marks: 3]

5 **(a)** $6x - 3 > 4x + 7$ **[M1]**

$2x > 10$ leading to $x > 5$ **[A1]**

(b) Open circle placed at 5 with an arrow pointing to the right **[B1]**

(c) 6 **[B1]**

[Total Marks: 4]

6 **(a)** $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) $-1, 0, 1, 2, 3, 4$ **[B2]**

(or [B1] for 5 correct with no incorrect, or all 6 correct with one incorrect)

[Total Marks: 4]

7 **(a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 and closed circle at 4 **[B1]**

Thick line joining the two circles **[B1]**

[Total Marks: 4]

8 **(a)** $4x \leq 20$ **[M1]**

$x \leq 5$ **[A1]**

(b) $3x + 6 < 20$ or $x + 2 < 6.66\ldots$ **[M1]**

4 **[A1]**

[Total Marks: 4]

9 **(a)** $2x < 12$ **[M1]**

$x < 6$ **[A1]**

(b) Recognises open circle means strict inequality ($>$ or $<$) and arrow to right means greater than **[M1]**

$x > -2$ **[A1]**

[Total Marks: 4]

10 **(a)** M1 for a correct first step, e.g., adding 1 to all parts to get $-4 < 2x \leq 8$

M1 for a correct second step, e.g., dividing all parts by 2

$-2 < x \leq 4$ **[A1]**

(b) M1 for an open circle at -2 and a closed solid circle at 4

A1 for a continuous thick line drawn between the two circles

[Total Marks: 5]

11 **(a)** Expands brackets correctly: $3x - 6 \leq 5x + 8$ **[M1]**

Rearranges to isolate x, e.g. $-14 \leq 2x$ or $-2x \leq 14$ **[M1]**

$x \geq -7$ **[A1]**

(b) Places a closed, solid circle at -7 **[B1]**

Draws a line with an arrow pointing to the right from the circle **[B1]**

[Total Marks: 5]

12 **(a)** Adds 1 to all three parts: $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 and closed (filled) circle at 4 **[B1]**

Horizontal line segment connecting the circles at -2 and 4 **[B1]**

(c) -1 **[B1]**

[Total Marks: 5]

13 **(a)** $-9 > 2x$ or $-2x > 9$ **[M1]**

$x < -4.5$ **[A1]**

(b) $-4 < 2n \leq 4$ **[M1]**

$-2 < n \leq 2$ **[M1]**

$-1, 0, 1, 2$ **[A1]**

(c) $x > -1$ (accept any valid variable) **[B1]**

[Total Marks: 6]

14 **(a)** $-15 \leq 3x$ or $-3x \leq 15$ **[M1]**

$x \geq -5$ or $-5 \leq x$ **[A1]**

(b) $-7 < 2x \leq 8$ **[M1]**

$-3.5 < x \leq 4$ **[A1]**

(c) Largest $= 4$ **[B1]**

Smallest $= -3$ **[B1]**

[Total Marks: 6]

15 **(a)** $-3x < 9$ or $2 - 11 < 3x$ **[M1]**

$x > -3$ **[A1]**

(b) $8x - 4 \leq 5x + 17$ **[M1]**

$3x \leq 21$ **[M1]**

$x \leq 7$ **[A1]**

(c) $-3 < n \leq 7$ **[M1]**

$-2, -1, 0, 1, 2, 3, 4, 5, 6, 7$ **[A1]**

[Total Marks: 7]

16 **(a)** Subtracts $3x$ and 3 to reach $\pm 4x \leq \pm 8$ or similar **[M1]**

$x \geq -2$ **[A1]**

(b) Solid circle (or equivalent clear inclusive mark) placed at -2 **[B1]**

Thick line or arrow pointing to the right starting from -2 **[B1]**

(c) Adds 5 to all parts to get $3 < 3x \leq 9$ **[M1]**

Divides by 3 to get $1 < x \leq 3$ **[A1]**

$2, 3$ **[A1]**

[Total Marks: 7]

17 $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Total Marks: 3]

18 $y \leq 5x - 4$ or equivalent **[B1]**

$2y \geq x + 1$ or equivalent **[B1]**

$x + y \leq 8$ or equivalent **[B1]**

[Total Marks: 3]

19 $x \geq 2$ **[B1]**

$y > x$ **[B1]**

$x + y \leq 8$ (or $y \leq -x + 8$) **[B1]**

[Total Marks: 3]

20 $x \geq 1$

[B1]

$y \geq 2$

[B1]

$x + y < 6$ (or $y < -x + 6$)

[B1]

[Total Marks: 3]

21 (a) $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 **[B1]**

Closed circle at 4 and a line joining the two circles **[B1]**

[Total Marks: 4]

22 (a) $-6 < 2x \leq 6$ (Allow M1 for one correct side e.g. $-6 < 2x$ or $2x \leq 6$) **[M2]**

$-3 < x \leq 3$ **[A1]**

(b) Open circle at -3 and closed circle at 3 **[B1]**

A single continuous line drawn connecting the two circles **[B1]**

[Total Marks: 5]

23 (a) (a) Adds 1 to all three parts: $-4 \leq 2x < 7$ **[M1]**

Divides all three parts by 2 **[M1]**

$-2 \leq x < 3.5$ or $-2 \leq x < \frac{7}{2}$ **[A1]**

(b) Award [B1] for at least three correct integers and no incorrect integers, or all correct integers with only one incorrect integer

$-2, -1, 0, 1, 2, 3$ **[B2]**

[Total Marks: 5]

24 (a) $x < 4$ **[B1]**

$y \geq -x + 3$ **[B1]**

$y \leq \frac{1}{2}x + 3$ (or equivalent, e.g. $2y \leq x + 6$) **[B1]**

(b) Substitutes $x = 3$ into the inequalities: $b \geq -3 + 3 = 0$ and $b \leq 0.5(3) + 3 = 4.5$ **[M1]**

Deduces the range $0 \leq b \leq 4.5$ **[A1]**

$0, 1, 2, 3, 4$ **[A1]**

[Total Marks: 6]

25 (a) $y \geq 0$ **[B1]**

Equation of sloping boundary identified as $x + y = 6$ or $y = -x + 6$ **[B1]**

$x + y < 6$ or $y < -x + 6$ (correct strict inequality symbol) **[B2]**

(b) Identifies valid integer coordinate $(1, 4)$ (or tests boundary lattice points such as $(1, 4), (2, 3), (3, 2)$) **[M1]**

14 **[A1]**

[Total Marks: 6]

26 $x \geq 1$ **[B1]**

$y \geq 2$ **[B1]**

$x + y < 6$ or $y < -x + 6$ **[B1]**

[Total Marks: 3]

27 (a) $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ or $y \leq -x + 6$ **[B1]**

(b) Tests a valid integer coordinate close to the boundary, e.g., $(5, 1)$, $(4, 2)$, or $(3, 3)$ **[M1]**

17 (from $x = 5, y = 1$) **[A1]**

[Total Marks: 5]

28 Identifies horizontal line $y = 2$ **[M1]**

States $y \geq 2$ (accept $y > 2$) **[A1]**

Identifies dashed line passing through $(0, 6)$ and $(4, 2)$ **[M1]**

Finds equation $y = -x + 6$ and states $y < -x + 6$ **[A1]**

Identifies solid line passing through $(0, 2)$ and $(1, 5)$ **[M1]**

Finds equation $y = 3x + 2$ and states $y \leq 3x + 2$ **[A1]**

[Total Marks: 6]

29 (a) Multiplies by 15 to clear denominators: $3(3x - 2) - 5(2x - 5) \leq 30$ **[M1]**

$9x - 6 - 10x + 25 \leq 30$ **[M1]**

$-x + 19 \leq 30$ **[M1]**

$x \geq -11$ **[A1]**

(b) Identifies that y can only take integer values of 2 or 3 (since $2y > x + y > 2 \implies y > 1$) **[M1]**

Finds correct points for one of the valid y values: e.g. $(1, 2)$ for $y = 2$ OR $(0, 3), (1, 3), (2, 3)$ for $y = 3$ **[B1]**

States all four correct points: $(1, 2), (0, 3), (1, 3), (2, 3)$ with no extras **[A1]**

[Total Marks: 7]

30 (a) Rearranging terms to get $3x \leq 15$ **[M1]**

$x \leq 5$ **[A1]**

(b) $1, 2, 3, 4, 5$ **[B2]**

If 0 scored, award 1 mark for $0, 1, 2, 3, 4, 5$ OR for omitting exactly one correct value **[B1]**

[Total Marks: 4]

31 $y \geq 2$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Total Marks: 3]

32 $x \geq 1$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

(Note: Accept strict inequalities $<, >$)

[Total Marks: 3]

33 **(a)** Subtracting 2 from all parts: $-9 < 3x \leq 9$ **[M1]**

Dividing by 3: $-3 < x$ and $x \leq 3$ **[M1]**

$-3 < x \leq 3$ **[A1]**

(b) 3 **[B1]**

[Total Marks: 4]

34 **(a)** Adding 1 to all parts: $-6 \leq 3x < 12$ **[M2, M1 for one side correct]**

Dividing by 3: $-2 \leq x < 4$ **[A1]**

(b) Identifies at least 3 correct integers or includes 4 but excludes -2 **[M1]**

$-2, -1, 0, 1, 2, 3$ **[A1]**

[Total Marks: 5]

35 **(a)** $y \geq 1$ **[B1]**

$y \leq x + 1$ **[B1]**

$y < 5 - x$ (or $x + y < 5$) **[B1]**

All three inequality signs correct (with strict inequality for the dashed line) **[B1]**

(b) Identifies $(1, 2)$ as a valid integer coordinate point inside R **[M1]**

$(1, 2)$ **[A1]**

[Total Marks: 6]

36 **(a)** Subtracts 1 from all parts: $-6 < 2x \leq 10$ **[M1]**

Divides by 2 across all parts **[M1]**

$-3 < x \leq 5$ **[A1]**

(b) Identifies at least two correct integers from the set **[M1]**

$-2, -1, 0$ **[A1]**

(c) Finds the correct roots of the quadratic $2x^2 - 7x + 3 = 0$ as 0.5 and 3 **[B1]**

$x < 0.5$ or $x > 3$ **[B1]**

[Total Marks: 7]

37 **(a)** $y \geq 1$ **[B1]**

$y \leq 2x - 2$ **[B2]**

(or $M1$ for finding the boundary equation $y = 2x - 2$)

$x + y \leq 7$ (or $y \leq -x + 7$) **[B2]**

(or $M1$ for finding the boundary equation $x + y = 7$)

(b) Identifies a valid integer coordinate inside or on the boundary of R that maximizes the expression, e.g., $(6, 1)$, $(5, 2)$, or $(4, 3)$ **[M1]**

13 (from $2(6) + 1 = 13$) **[A1]**

[Total Marks: 7]

38 **(a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) $-1, 0, 1, 2, 3, 4$ **[B2]**

1 omission or 1 extra integer **[B1]**

(c) **(i)** $2(3x + 2) + 2x < 40$ or $(3x + 2) + x < 20$ **[M1]**

$8x + 4 < 40$

8x ¡ 36

x ¡ $4.5 with outer errors [A1]$

(ii) Identifies maximum integer $x = 4$ **[M1]**

Length = 14, Width = 4, Area = 56 **[A1]**

[Total Marks: 8]

39 **(a)** $y \geq 0.5x + 0.5$ or $2y \geq x + 1$ (Accept equivalent solid boundary) **[B1]**

$y \leq 6x - 5$ (Accept equivalent solid boundary) **[B1]**

$y < -\frac{4}{3}x + \frac{29}{3}$ or $4x + 3y < 29$ (Must be strict inequality for dashed line) **[B1]**

(b) Identifies a valid integer point near the boundary that maximizes the sum, e.g., $(2, 6), (3, 5)$, or $(4, 4)$. Note: $(2, 7)$ is invalid as it lies on the dashed boundary **[M1]**

8 **[A1]**

[Total Marks: 5]

40 **(a)** $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y < 8$ **[B1]**

(b) Identifies $(4, 3)$ or $(5, 2)$ as integer coordinates within the region R. Note that $(4, 4)$ is excluded due to the strict inequality $x + y < 8$. **[M1]**

Evaluates $2x + 3y$ and correctly identifies $(4, 3)$ as yielding the maximum value (17). **[A1]**

[Total Marks: 5]

41 **(a)** $x \geq 15$ **[B1]**

$x + y \leq 40$ **[B1]**

$y \geq \frac{1}{2}x$ or $2y \geq x$ **[B1]**

(b) Identifies the objective function for profit: $P = 400x + 150y$ **[M1]**

Tests the valid integer vertex $(26, 14)$ or evaluates the profit equation showing profit is maximised at the highest possible integer value for x **[M1]**

12500 **[A1]**

[Total Marks: 6]

42 **(a)** $x \geq 2$ **[B1]**

$y \geq 1$ **[B1]**

$x + y < 7$ or $y < -x + 7$ **[B1]**

(b) Finding critical values -0.45 and 4.45 (allow ≈ -0.449 and ≈ 4.449) **[B2]**

(Allow B1 for setting $x^2 - 4x - 2 = 0$ and finding one correct root or correct quadratic formula setup)

$-0.45 \leq x \leq 4.45$ **[B1]**

[Total Marks: 6]

43 **(a)** Draw correct solid line $y = 2$

[B1]

Draw correct solid line $y = x + 2$

[B1]

Draw correct solid line $2x + y = 8$

[B1]

Correct shading of unwanted regions, leaving the triangle with vertices $(0, 2)$, $(3, 2)$, and $(2, 4)$ unshaded and clearly representing R.

[B1]

(b) Identify integer coordinates $(2, 4)$ or $(3, 2)$ as yielding the maximum value

[M1]

14

[A1]

[Total Marks: 6]

44 **(a)** $2[(2x + 5) + (x - 1)] < 50$ or $6x + 8 < 50$ **[M1]**

$6x < 42 \implies x < 7$ **[A1]**

(b) $(2x + 5)(x - 1) \geq 42$ **[M1]**

$2x^2 + 3x - 5 \geq 42 \implies 2x^2 + 3x - 47 \geq 0$ **[A1]**

(c) Finds positive root of $2x^2 + 3x - 47 = 0$ as 4.155... (4.16) **[M1]**

Solves quadratic inequality for $x > 1$: $x \geq 4.16$ **[A1]**

$4.16 \leq x < 7$ **[A1]**

[Total Marks: 7]

45 (a) $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 8$ **[B1]**

(b) Identifies at least two valid integer coordinate pairs within the region R (e.g., $(6, 2)$, $(5, 3)$, $(4, 4)$) **[M1]**

Evaluates $2x + 3y$ for at least one critical integer point, e.g., $2(4) + 3(4) = 20$ or $2(5) + 3(3) = 19$ **[M1]**

20 **[A1]**

(c) Identifies the correct total number of integer points in R is 9, or identifies the 3 correct points where $x = y$: $(2, 2), (3, 3), (4, 4)$ **[M1]**

$\frac{3}{9}$ or $\frac{1}{3}$ **[A1]**

[Total Marks: 8]

46 (a) $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Multiplies throughout by 12 (or uses common denominator): $4(2x - 5) - 3(x - 3) > 12$ **[M1]**

Expands and simplifies: $8x - 20 - 3x + 9 > 12 \implies 5x - 11 > 12 \implies 5x > 23$ **[M1]**

5 **[A1]**

(c) Constructs correct inequality: $3.20 + 1.50k > 5.00 + 1.10k$ **[M1]**

Solves for k: $0.40k > 1.80 \implies k > 4.5$ **[M1]**

5 **[A1]**

[Total Marks: 8]

47 (a) $x + y \leq 500$ **[B1]**

$y \geq 150$ **[B1]**

$5x + 10y \geq 3000$ (or simplified to $x + 2y \geq 600$) **[B1]**

(b) Line drawn correctly for $x + y = 500$ (passing through $(0, 500)$ and $(500, 0)$) **[B1]**

Line drawn correctly for $y = 150$ (horizontal line) **[B1]**

Line drawn correctly for $x + 2y = 600$ (passing through $(0, 300)$ and $(600, 0)$) **[B1]**

Correct region identified and all unwanted regions properly shaded (leaving quadrilateral with vertices $(0, 300), (0, 500), (350, 150), (300, 150)$ unshaded) **[B1]**

(c) Identifies the vertex $(350, 150)$ as giving the maximum x value **[M1]**

Children's tickets = 350, Adult tickets = 150 **[A1]**

[Total Marks: 9]

48 (a) $x + y \leq 60$ **[B1]**

$x \geq 10$ **[B1]**

$y \geq 0.5x$ (or $2y \geq x$ or $y \geq \frac{1}{2}x$) **[B1]**

(b) Draws the line $x + y = 60$ correctly (from $(0, 60)$ to $(60, 0)$) **[B1]**

Draws the line $x = 10$ correctly (vertical line at 10) **[B1]**

Draws the line $y = 0.5x$ correctly (from $(0, 0)$ passing through $(20, 10)$, $(40, 20)$, etc.) **[B1]**

Shades the unwanted regions correctly, leaving the correct triangular region R unshaded **[B1]**

(c) Identifies $(40, 20)$ as the vertex for maximum profit (or tests multiple vertices) **[M1]**

2100 **[A1]**

[Total Marks: 9]

49 (a) $x + y \leq 60$ **[B1]**

$x + 2y \leq 80$ **[B1]**

$y \geq 10$ **[B1]**

(b) B1 for correctly drawing the line $x + y = 60$

B1 for correctly drawing the line $x + 2y = 80$

B1 for correctly drawing the line $y = 10$

B1 for correctly shading the unwanted regions (above the two diagonal lines, and below the horizontal line) to leave the correct valid region unshaded

(c) M1 for a valid method to find the maximum profit, e.g., testing the vertex $(40, 20)$ in the objective function $5x + 8y$

360 **[A1]**

[Total Marks: 9]

50 (a) Forms left inequality: $3(2x - 3) < 4(x + 5)$ or equivalent **[M1]**

Simplifies to $6x - 9 < 4x + 20$

x ¡ $\frac{29}{2}$ (or 14.5) [A1]

Forms right inequality: $2(x+5) \leq 3(3x-1)$ or equivalent **[M1]**

Simplifies to $2x + 10 \leq 9x - 3$

x $\geq \frac{13}{7}$ [A1]

$\frac{13}{7} \leq x < \frac{29}{2}$ (accept $1.86 \leq x < 14.5$ or equivalent double inequality) **[A1]**

(b) (i) $25 < 12 + 0.07x$ (or $12 + 0.07x > 25$) **[B1]**

(ii) $13 < 0.07x$ **[M1]**

$x > 185.71\ldots$ **[A1]**

186 **[A1]**

[Total Marks: 9]

51 (a) $x + y \leq 60$ **[B1]**

$y \geq 10$ **[B1]**

$x \geq 15$ **[B1]**

$y \leq 2x$ **[B1]**

(b) Line $x + y = 60$ drawn correctly **[B1]**

Line $y = 10$ drawn correctly **[B1]**

Line $x = 15$ drawn correctly **[B1]**

Line $y = 2x$ drawn correctly **[B1]**

Correct region R indicated with all four unwanted regions shaded **[B1]**

(c) Identifies $(20, 40)$ as the vertex for maximum profit **[M1]**

Mountain bicycles = 20 and Road bicycles = 40 **[A1]**

Maximum profit = 2600 **[A1]**

[Total Marks: 12]